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Q1 Prove that a Pedersen Commitment is homomorphic

Q2 How is the private key SK of an user named “bob” constructed in the Boneh-Franklin’s Identity Based Encryption scheme? (notation: $s, g^s = \text{PKG}$ key pair, $H()$ = hash function which maps a string into an EC point)

- ☐ a) $SK = g^{H(\text{bob})}$
- ☒ b) $SK = H(\text{bob})^s$
- ☐ c) $SK = \text{bob}^s$
- ☐ d) $SK = g^s H(\text{bob})$

Q3 In ECDSA, the key pair (private key, public key) is...

- ☐ a) A pair of EC points
- ☐ b) A pair of modular integers
- ☒ c) the private key is a modular integer whereas the public key is an EC point
- ☐ d) the private key is an EC point whereas the public key is a modular integer

Q4 A Secret Sharing scheme is ideal if...

- ☐ a) Each party receives exactly one share
- ☐ b) The total number of participating parties n is equal to the minimum number of parties t which can reconstruct the secret
- ☐ c) the size of each share is an integer value
- ☒ d) none of the above answers

Q5 Describe the RSA common modulus attack

Q6 Determine the access control matrix that implements the policy: $P = \mathbf{A} \text{ AND } \mathbf{B} \text{ AND } (\mathbf{C} \text{ OR } (\mathbf{D} \text{ AND } \mathbf{E}))$

A:	1	-1	-1	0
B:	0	1	0	0
C:	0	0	1	0
D:	0	0	1	-1
E:	0	0	0	1

(solution obviously not unique!)

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E1 Consider the Elliptic curve $y^2 = x^3 + x + 1$ defined over the modular integer field Z_7 .

A. find all the points $EC(Z_7)$

$$P = (x_1, y_1)$$

$$Q = (x_2, y_2)$$

$$R = P + Q = (x_3, y_3)$$

$$x_3 = \lambda^2 - x_1 - x_2$$

$$y_3 = \lambda(x_1 - x_3) - y_1$$

B. State what is the order of the corresponding group

$$\lambda = \begin{cases} \frac{y_2 - y_1}{x_2 - x_1} & P \neq Q \\ \frac{3x_1^2 + a}{2y_1} & P = Q \end{cases}$$

C. Compute $[3](2,2)$

[HELP: possibly useful mnemonic hints reported here on the right;

MUST-DO: show step-by-step detailed computations]

points: 0, (0,1), (0,6), (2,2), (2,5)

order: 5

$[3](2,2) = (0,6) \rightarrow$ should be computed as follows:

$$(2,2) + (2,2) = (0,1)$$

$$(2,2) + (0,1) = (0,6)$$

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E2 Assume arithmetic modulus 101. A Linear secret sharing scheme involving 4 parties is described by the following access control matrix:

A:	1	1	0
B:	0	1	-1
C:	0	0	-1
D:	0	1	1

A. Assume that the following shares are revealed:

A → 23

B → 88

C → 57

What is the secret? (explain how you arrived to the result, otherwise the answer is not considered valid)

$$93 = 23 - 88 + 57$$

B. [optional, extra] Assume that the following shares are revealed:

A → 79

B → 20

D → 7

What is the secret? (explain how you arrived to the result, otherwise the answer is not considered valid)

Per ricostruire il vettore (1,0,0) è necessario fare la seguente operazione:

$A - (B + D)/2$ ma attenzione che l'aritmetica è modulare!! Pertanto $\frac{1}{2} = 51 \pmod{101}$ e quindi

$$\text{Segreto} = 79 - 27 \times 51 \pmod{101} = 15$$

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E3 – part 1 – El Gamal Encryption, $g=29$, $p=83$:

1. Reviewed El Gamal encryption

Ciphertext = $\{g^r, m \cdot h^r\}$

2. Assume operations are modulo $p=83$: is $g=29$ a generator of the Z_{83}^* multiplicative group? [you must respond to this question by performing a single “test”! Trying all possible values in the range is not considered a valid answer]

It suffices to compute $g^{(p-1)/2}$, since $29^{41} \bmod 83 = 1$, g is NOT a generator.

3. Using $g=29$ and $p=83$, encrypt message $M=37$ for an user whose private key is $sk=7$ and whose public key is $pk=4$ – if you need an ephemeral value, use $r=13$.

Ciphertext = $\{g^r, m \cdot pk^r\} \rightarrow$ using $r=13$, $pk=4$, $M=37 \rightarrow$ ciphertext = $\{12, 51\}$

E3 – part 2 – Threshold El Gamal Decryption.

If you have not solved the previous part, solve the exercise by using as ciphertext the pair $\{41, 25\}$ [note: on purpose different from the solution of the previous exercise!]

The ciphertext produced at the end of the previous part is now sent for threshold decryption to a $(2,3)$ group. The group has been built by sharing the secret key via a $(2,3)$ Shamir Secret Sharing scheme, prime modulus 41.

The three participating parties P_1, P_2, P_3 , use standard x-coordinates $x_i = \{1, 2, 3\}$.

The message is received by parties P_1 and P_3 which have shares $s_1=26$ and $s_3=23$, respectively

- compute the Lagrange interpolation coefficients for parties 1 and 3;

$q=41$; $x_1=1$; $x_3=3$;

$\lambda_1 = \text{Mod}[-x_3 * \text{PowerMod}[x_1 - x_3, -1, q], q] = 22$

$\lambda_3 = \text{Mod}[-x_1 * \text{PowerMod}[x_3 - x_1, -1, q], q] = 20$

- Assuming that P_1 and P_3 directly exchange their shares, reconstruct the original secret key

$s_1=26$; $s_3=23$;

$\text{Mod}[s_1 * \lambda_1 + s_3 * \lambda_3, q] = 22 \times 26 + 20 \times 23 \bmod 41 = 7$

- Assuming, instead, that P_1 and P_3 do NOT explicitly exchange their shares: show how P_1 and P_3 can still cooperate to decrypt the previous El Gamal encrypted message (and numerically compute the result, showing the step-by-step operations).

start from $\{g^r, m \cdot h^r\} = \{12, 51\}$.

P_1 computes $12^{s_1 \cdot \lambda_1} \bmod 83 = 49$;

P_3 computes $12^{s_3 \cdot \lambda_2} \bmod 83 = 28$;

Now multiply the two terms and compute the modular inverse $\rightarrow (49 \cdot 28)^{(-1)} = 17$

And decrypt the message as $17 \cdot 51 \bmod 83 = 37$

Network Security – prof. Giuseppe Bianchi – 3rd term exam, 4 February 2021

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